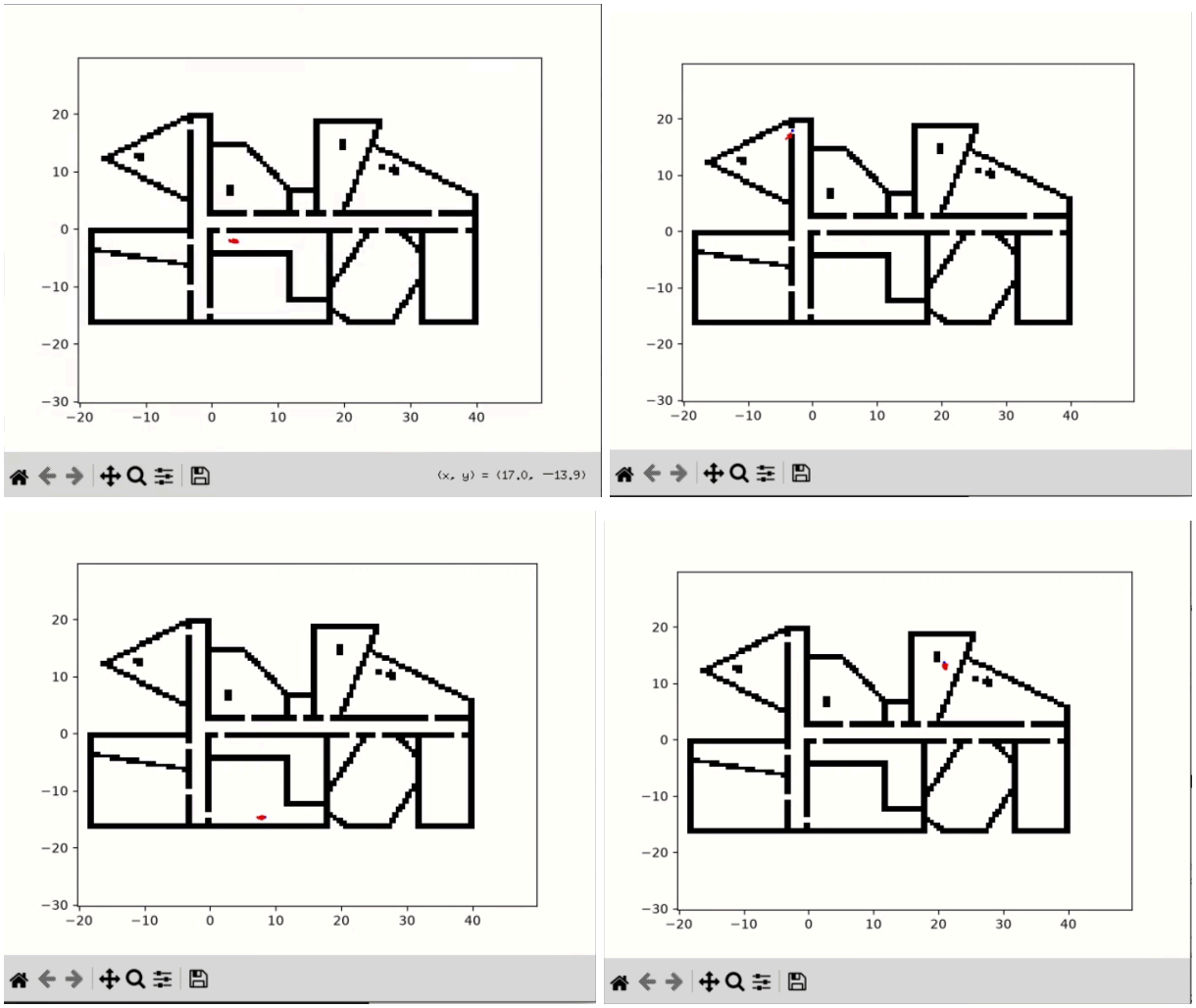
Lab 3: Particle Filter Localization
Austin J. Erickson

Deliverables for lab 3:

2.4:
TODO: Include several images showing your algorithm's output at different time steps



TODO: Include the specific parameter values you selected in your writeup

Parameters used for the above images:

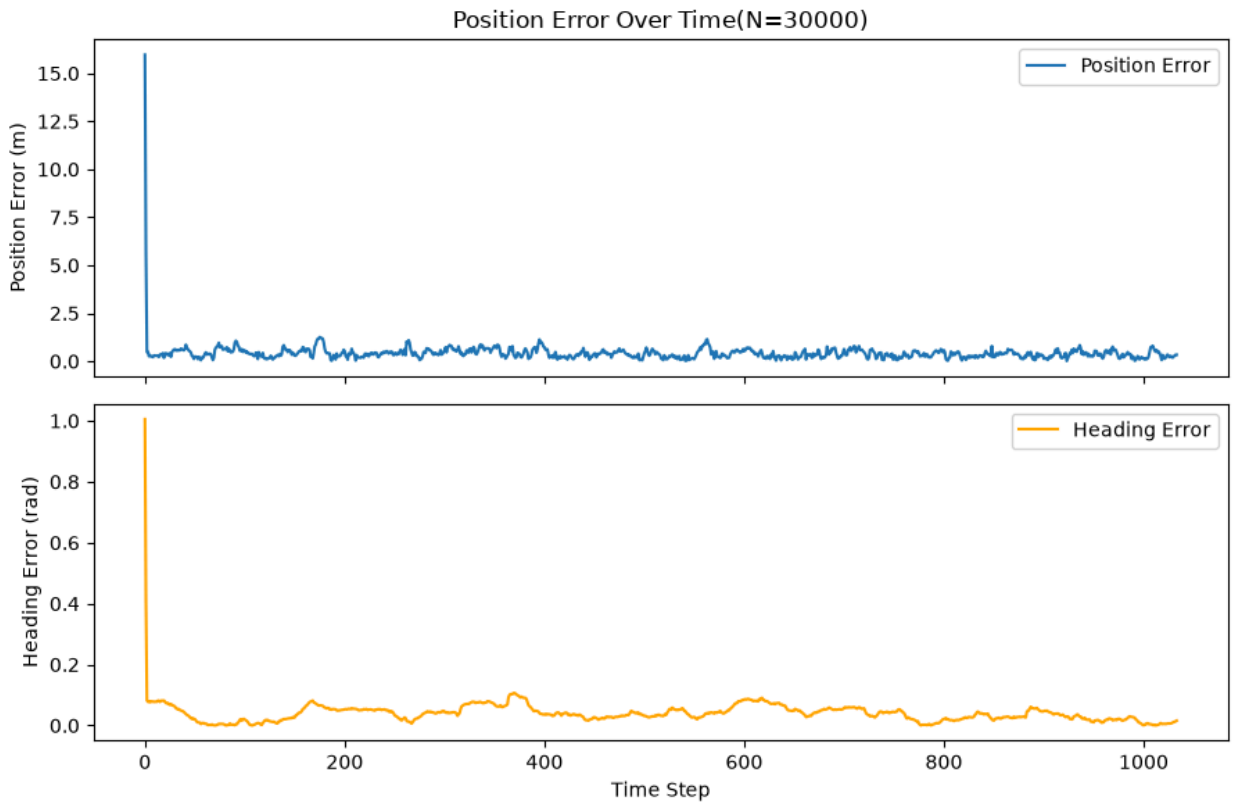
```
alphas = np.array([0.8, 0.0, 0.05, 0.15])
   cov = np.array([.04, .04, .01])
```

2.5: Evaluation

TODO: Remove the jumpstart particle, and test how many particles it takes to properly localize. This will slow down your program, so feel free to kill it after the first couple of time steps. Include the number of particles in your writeup.

It took *many* particles to properly localize. While I could achieve localization on particle counts as low as 5000, that was after multiple different failed tries. The lowest value I found consistently was able to work was around 25000-30000, though those occasionally had failed runs. To truly achieve reliable localization, I'd estimate you'd want upwards of 50,000 particles, though I did not want to put my computer through that. Many of the errors were due to the localization assuming the particle was at the wrong door early on (due to the hallway being very similar along its length), which caused errors for the rest of the run.

TODO: Generate two plots showing the error between the mean position/rotation of the particles and the true values stored in the X_t vector, versus time. This error should shrink to zero as your lter converges to the true solution.



2.6: Testing

TODO: Include the output of 'pytest -v -rN --tb=no --no-header' in your writeup.

```
(base) austiner99@radlab15:~/classes/localization_and_mapping_f26/coding_ass_3$ pytest -v
-rN --tb=no --no-header
========================================== test session starts
==========================================
collected 14 items

tests/test_expected_measurement.py::test[ans0] PASSED                    [  7%]
tests/test_expected_measurement.py::test[ans1] PASSED                    [ 14%]
tests/test_expected_measurement.py::test[ans2] PASSED                    [ 21%]
tests/test_expected_measurement.py::test[ans3] PASSED                    [ 28%]
tests/test_expected_measurement.py::test[ans4] PASSED                    [ 35%]
tests/test_expected_measurement.py::test[ans5] PASSED                    [ 42%]
tests/test_expected_measurement.py::test[ans6] PASSED                    [ 50%]
tests/test_iterate.py::test PASSED                                       [ 57%]
tests/test_normalize_weights.py::test PASSED                             [ 64%]
tests/test_propagate_motion.py::test PASSED                              [ 71%]
tests/test_resample_particles.py::test PASSED                            [ 78%]
tests/test_update_weight.py::test[ans0] PASSED                           [ 85%]
tests/test_update_weight.py::test[ans1] PASSED                           [ 92%]
tests/test_update_weight.py::test[ans2] PASSED                           [100%]

========================================== 14 passed in 1.12s
==========================================
```